

# Het Patel

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## Education

University at Buffalo, State University of New York

August 2023 - May 2027

B.S. in Computer Science

B.A. in Statistics

- **GPA:** 3.96/4.0
- **Coursework:** Operating Systems, Distributed Systems, Reinforcement Learning, Machine Learning, Computer Vision, Compilers, Stochastic Processes, Regression

## Experience

Full-Stack Software Developer Intern

August 2025 - October 2025

Athlynk Inc.

Omaha, NE (Remote)

- Developed mobile application using React Native and RESTful APIs with FastAPI on the server.
- Implemented messaging service with WebSockets. Optimized database CRUD operations with PostgreSQL.

Undergraduate Teaching Assistant

January 2026 - Present

Department of Computer Science @ UB

Buffalo, NY

- For Machine Learning, covered topics such as the supervised and unsupervised learning, decision trees, neural networks, SVMs, and reinforcement learning.
- For Computer Organization, assisted with MIPS assembly, datapath design, pipelining, memory hierarchy, and ALU.

## Research

Research Assistant

May 2026 - Present

Dr. Roshan Ayyalasomayajula

[WIRES Lab](#) 🔗

- Developed a privacy preserving ML-based wireless localization framework which guarantees Differential Privacy.
- Designed a Split Neural Network trained using DP-SGD and Federated Learning which predicts device location using WiFi signals.

Research Assistant

May 2026 - Present

Dr. Wen Yao Xu

[ESC Group](#) 🔗

- Contributed to mRehab, a mobile rehabilitation app that helps post-stroke patients rebuild mobility and independence.
- Built a React Native application which uses built-in motion sensors (accelerometer and gyroscope) to track movements.

## Technologies

**Languages:** C, Go, Python, Java, Kotlin, OCaml, JavaScript, TypeScript

**Frameworks:** React Native, Express.js, TensorFlow, Pytorch, scikit-learn, OpenCV, Opacus

**Database:** PostgreSQL, Supabase

**Tools:** Git, Docker, gdb, Expo, CometML, Minitab

## Projects

CrowdSense

[demo](#) 🔗

**Tools:** Ultralytics YOLO, Pytorch, OpenCV, FastAPI, Websockets, React Native, Expo

- Built a real-time crowd monitoring system with Ultralytics YOLO model to detect occupancy level from camera feed.
- Designed a multi-service architecture integrating a Pytorch model pipeline, FastAPI server, and React Native mobile app using WebSockets.

Pintos Operating System (Stanford)

[github](#) 🔗

**Tools:** C, gdb, Makefile

- Implemented threading component of the OS kernel including synchronization primitives and Multilevel Feedback Queue Scheduler (MLFQS).
- Built system call interface including argument validation, safe user memory access, and kernel-user mode transitions, system calls, and process control logic.